

Mean, Median, Mode — Explained Simply

Three ways to describe a dataset with ONE number

Real banking example: We'll use daily ATM withdrawals. Let's go!

THE DATA

ATM withdrawals on a particular day (10 customers):

₹5,000 | ₹10,000 | ₹3,000 | ₹15,000 | ₹8,000 | ₹5,000 | ₹12,000 | ₹5,000 | ₹100,000 | ₹6,000

When sorted (smallest to largest):

₹3,000 | ₹5,000 | ₹5,000 | ₹5,000 | ₹6,000 | ₹8,000 | ₹10,000 | ₹12,000 | ₹15,000 | ₹100,000

Notice: One outlier (₹100,000 withdrawal by a business owner). Rest are normal customers.

1 MEAN (Average)

Definition:

Sum all values and divide by count.

Formula:

$\text{Mean} = (5000 + 10000 + 3000 + 15000 + 8000 + 5000 + 12000 + 5000 + 100000 + 6000) / 10$
 $\text{Mean} = 169000 / 10$
 $\text{Mean} = ₹16,900$

What it means:

Average withdrawal is ₹16,900

⚠ BUT LOOK:

Only 1 customer withdrew ₹100,000 (business owner).

Most customers withdrew ₹3,000 to ₹15,000.

So the mean of ₹16,900 is misleading! It's pulled UP by the one outlier.

When to use Mean:

- ✓ Data has no outliers (normal distribution)
- ✓ You want total sum divided equally (fair sharing)
- ✓ Example: 10 employees earned ₹10K, ₹12K, ₹15K, ₹11K, ₹13K each → mean = ₹12.2K (fair salary average)

2 MEDIAN (Middle Value)

Definition:

The middle value when data is sorted. 50% above, 50% below.

How to find it:

Sort data: ₹3K, ₹5K, ₹5K, ₹5K, ₹6K, ₹8K, ₹10K, ₹12K, ₹15K, ₹100K

Count = 10 (even number)

Take middle two: position 5 (₹6K) and position 6 (₹8K)

Median = $(₹6K + ₹8K) / 2 = ₹7K$

What it means:

The typical ATM withdrawal is ₹7,000

Half customers withdrew less than ₹7K, half more than ₹7K

✅ Why Median is better here:

The ₹100,000 outlier didn't affect it!

Median (₹7K) is more representative of typical customer than Mean (₹16.9K)

When to use Median:

- ✅ Data has outliers (e.g., one billionaire in 100 normal people)
- ✅ You want the "typical" or "middle" value
- ✅ Example: House prices. One ₹10 crore mansion pulls mean up. Median is more real.

3 MODE (Most Frequent)

Definition:

The value that appears most often.

How to find it:

Count frequency of each value:

Amount	Frequency
₹3,000	1 time

Amount	Frequency
₹5,000	3 times ← MOST FREQUENT
₹6,000	1 time
₹8,000	1 time
₹10,000	1 time

Mode = ₹5,000 (appears 3 times)

What it means:

The most common withdrawal amount is ₹5,000
If a bank wants to stock enough ₹500 notes, knowing mode helps.

When to use Mode:

- ✔ Categorical data (sizes: S, M, L, L, L, XL → mode = L)
- ✔ You want the "most popular" value
- ✔ Example: Most common shoe size sold → stock more of that size
- ✔ Example: Most common account type among customers

COMPARISON TABLE

Aspect	Mean	Median	Mode	Value
Definition	Average of all values	Middle value when sorted	Most frequently occurring value	
Formula	Sum ÷ Count	Middle of sorted data	Highest frequency	
Our example	₹16,900	₹7,000	₹5,000	
Affected by outliers?	✔ YES (big effect)	✗ NO (immune)	✗ NO (immune)	
Use when	Data is normal, no outliers	Outliers exist, want "typical"	Want "most popular" value	
Example	Salary average	House prices, income	Shoe size, color preference	

MORE REAL BANKING EXAMPLES

Example 1: Loan amounts (in ₹ lakhs)

Data: 10, 15, 12, 18, 14, 12, 20, 12, 100, 13

Mean = $(10+15+12+18+14+12+20+12+100+13) / 10 = 226 / 10 = ₹22.6$ lakhs

Median = sorted: 10, 12, 12, 12, 13, 14, 15, 18, 20, 100 $\rightarrow (13+14)/2 = ₹13.5$ lakhs

Mode = ₹12 lakhs (appears 3 times)

⚠ Notice: One customer took ₹100 lakhs (business loan) pulled mean way up. Median (₹13.5L) is more representative of typical personal loan.

Example 2: Credit card transaction amounts on a day

Data: ₹500, ₹2000, ₹500, ₹1500, ₹500, ₹1000, ₹800, ₹500, ₹600, ₹700

Mean = $₹8300 / 10 = ₹830$

Median = sorted: 500, 500, 500, 500, 600, 700, 800, 1000, 1500, 2000 $\rightarrow (500+600)/2 = ₹550$

Mode = ₹500 (appears 4 times)

💡 Most transactions are around ₹500. Mean of ₹830 is misleading.

Example 3: Account ages (in years)

Data: 1, 3, 2, 5, 2, 2, 8, 2, 15, 3

Mean = $43 / 10 = 4.3$ years

Median = sorted: 1, 2, 2, 2, 2, 3, 3, 5, 8, 15 $\rightarrow (2+3)/2 = 2.5$ years

Mode = 2 years (appears 4 times)

👁 Bank knows: typical customer has had account ~2.5 years. One very loyal customer (15 years) pulls mean up.

WHY THIS MATTERS FOR ML

Feature Engineering:

When cleaning data, you replace missing values with mean, median, or mode:

- Missing age? Use median (immune to outliers like billionaires)
- Missing category (account type)? Use mode (most common type)

Data Analysis:

Understanding distribution of features helps pick right metrics:

- Income has outliers (billionaires)? Use median instead of mean

- Transaction amounts? Mode tells you most common transaction size

In your fraud model:

Mean transaction amount = misleading (outliers pull it up)

Median transaction amount = better for understanding typical behavior

Mode helps identify most common transaction size for anomaly detection

QUICK REFERENCE — When to Use Each

Measure	Use When...	Example
MEAN	Data has no outliers, you want fair average	Employee salaries: ₹10K, ₹12K, ₹15K → mean = ₹12.3K
MEDIAN	Data has outliers, you want "typical" value	House prices: 10L, 15L, 20L, 500L (mansion) → median = 17.5L
MODE	You want "most common" or categorical data	Shoe sizes: S, M, L, L, L, XL → mode = L (stock more L)

Remember: Mean is sensitive, Median is robust, Mode is popular. Choose based on what you want to know!